

25CS1302E - DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND

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SECTION: 2

LAB COURSEWORK

Q. Week 1: LAB COURSEWORK A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow . As a database developer, your task is to create the SQL foundation for this system. First, create a new database named bookflow_db . Then, design two tables: a Books table containing the columns book_id , title , isbn , and published_year , and a Members table containing the columns member_id , full_name , and email . Ensure data integrity by setting book_id and member_id as PRIMARY KEYs , applying a NOT NULL constraint to the title column so that every book must have a name, enforcing UNIQUE constraints on both isbn and email to prevent duplicate entries, and adding a CHECK constraint to ensure that the published_year is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the Books and Members tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as CREATE DATABASE , CREATE TABLE , and INSERT statements

SOL:-

Enter password: ****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 11

Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE library_kavya;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> USE library _kavya;
```

Database changed

```
mysql> SELECT DATABASE();
```

```
+-----+
```

```
| DATABASE() |
```

```
+-----+
```

```
| library_kavya |
```

```
+-----+
```

1 row in set (0.00 sec)

```
mysql> CREATE TABLE books (
```

```
->  book_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
->  title VARCHAR(255) NOT NULL,
```

```
->  isbn VARCHAR(13) NOT NULL UNIQUE,
```

```
->  published_year INT,
```

```
->  CONSTRAINT chk_published_year CHECK (published_year < 2027)
```

```
-> );
```

Query OK, 0 rows affected (0.16 sec)

```
mysql> DESC books;
```

```
+-----+-----+-----+-----+-----+-----+
```

```
| Field      | Type      | Null | Key | Default | Extra      |
```

```
+-----+-----+-----+-----+-----+-----+
```

```
| book_id    | int       | NO   | PRI | NULL    | auto_increment |
```

```
| title      | varchar(255) | NO   |     | NULL    |                |
```

```
| isbn       | varchar(13) | NO   | UNI | NULL    |                |
```

```
| published_year | int       | YES  |     | NULL    |                |
```

```
+-----+-----+-----+-----+-----+-----+
```

4 rows in set (0.02 sec)

```
mysql> CREATE TABLE members (
```

```

-> member_id INT AUTO_INCREMENT PRIMARY KEY,
-> full_name VARCHAR(100) NOT NULL,
-> email VARCHAR(150) NOT NULL UNIQUE
-> );

```

Query OK, 0 rows affected (0.08 sec)

```
mysql> DESC members;
```

```

+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| member_id | int      | NO   | PRI | NULL    | auto_increment |
| full_name | varchar(100) | NO   |     | NULL    |              |
| email     | varchar(150) | NO   | UNI | NULL    |              |
+-----+-----+-----+-----+-----+-----+

```

3 rows in set (0.00 sec)

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
    -> VALUES ('Rich Dad Poor Dad', '9780446691262', 1997);
```

Query OK, 1 row affected (0.02 sec)

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
    -> VALUES ('Deep Work', '9781455586691', 2016);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
    -> VALUES ('The Psychology of Money', '9780857197689', 2020);
```

Query OK, 1 row affected (0.02 sec)

```
mysql> SELECT * FROM books;
```

```

+-----+-----+-----+-----+
| book_id | title                                | isbn      | published_year |
+-----+-----+-----+-----+

```

	1	Rich Dad Poor Dad	9780446691262		1997	
	2	Deep Work	9781455586691		2016	
	3	The Psychology of Money	9780857197689		2020	

+-----+-----+-----+-----+

3 rows in set (0.00 sec)

mysql> INSERT INTO members (full_name, email)

-> VALUES ('Aarav Sharma', 'aarav.sharma@gmail.com');

Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO members (full_name, email)

-> VALUES ('Meera Kapoor', 'meera.kapoor@gmail.com');

Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO members (full_name, email)

-> VALUES ('Rohan Patel', 'rohan.patel@gmail.com');

Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM members;

+-----+-----+-----+-----+

	member_id		full_name		email	
--	-----------	--	-----------	--	-------	--

+-----+-----+-----+-----+

	1		Aarav Sharma		aarav.sharma@gmail.com	
--	---	--	--------------	--	------------------------	--

	2		Meera Kapoor		meera.kapoor@gmail.com	
--	---	--	--------------	--	------------------------	--

	3		Rohan Patel		rohan.patel@gmail.com	
--	---	--	-------------	--	-----------------------	--

+-----+-----+-----+-----+

3 rows in set (0.00 sec)

mysql>